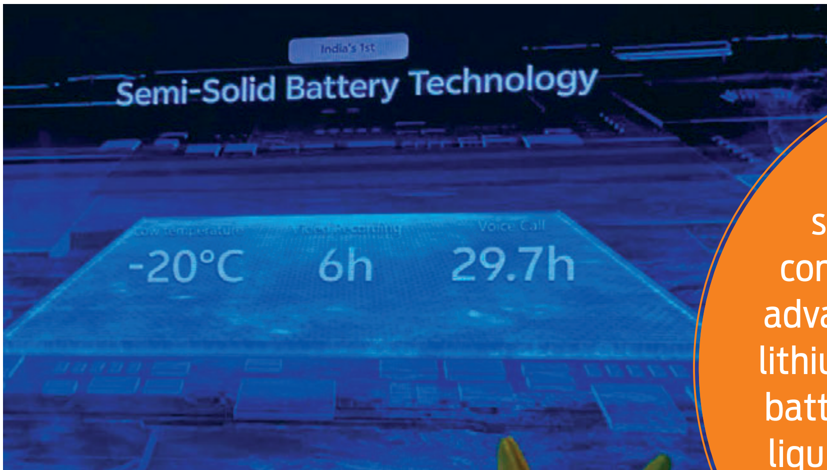

Semi-solid and solid-state batteries are often compared, as both represent advancements over traditional lithium-ion designs. Solid-state batteries completely eliminate liquid components, using solid electrolytes for even greater safety and energy density.

is less likely to leak or ignite, making these batteries safer for everyday use.

Improved Longevity: The semi-solid design reduces wear and tear on battery components, which translates to a longer lifespan. Vivo's implementation means users can enjoy consistent battery performance for years.

Semi-solid vs Solid-state Batteries

Semi-solid and solid-state batteries are often compared, as both represent advancements over traditional lithium-ion designs. Solid-state batteries completely eliminate liquid components, using solid electrolytes for even greater safety and energy density. However, they are currently expensive and challenging to mass-produce for consumer devices.

Semi-solid batteries offer a middle ground, combining some benefits of solid-state technology with the cost and scalability advantages of liquid-based designs.

What's Next: Solid-state batteries in smartphones

Solid-state batteries are heralded as the future of smartphone power. These batteries replace liquid electrolytes entirely with solid materials, resulting in higher energy densities, faster charging, and improved safety. Companies like Samsung are making significant strides in this domain, with prototypes showcasing promising results. In fact, Samsung claims that their solid-state batteries can charge from 10% to 80% capacity in nine minutes.

However, challenges like production costs and scalability mean widespread adoption is still years away. When they do arrive, solid-state batteries could revolutionise smartphones, making them lighter, safer, and more powerful.

Real-world performance of Vivo's Semi-solid Battery

For Indian consumers, Vivo's semi-solid battery technology marks a step forward. Imagine you're on a winter trek in Manali, capturing breathtaking landscapes. Traditional batteries might struggle with sudden shutdowns, but the X200 Pro promises to withstand the cold and deliver uninterrupted performance. The phone promises to be a treat for camera professionals and for those relying on their phones in harsh environments, this innovation offers some assurance.

In our review, we found the Vivo X200 Pro to deliver impressive battery performance. Its 6,000 mAh Silicon carbon battery lasted nearly 20 hours in the PCMark Battery Test, outperforming competitors like the OPPO Find X8 Pro and Realme GT 7 Pro.

In daily usage, the device provided 9-10 hours of screen-on time with medium to heavy use. Charging is efficient, with the 90W wired charger taking the battery from 0 to 100% in 45 minutes.